Std ID:_____ Name:_____ :Section:_____

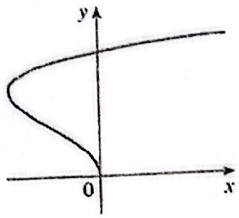
Quiz 1

Topic: *Parametric Equations*

Q.1 Sketch the curve by using the parametric equations to plot points. Indicate with an arrow the direction in which the curve is traced as increases $x = 1 + \sqrt{t}\,, y = t^2 - 4t\,; 0 \le t \le 5$

Q.2 Find the area enclosed by the given parametric curve and the y-axis.

$$x = t^2 - 2t\,,$$
$$y = \sqrt{t}$$



Quiz 1

Topic: *Parametric Equations*

Q.1 Eliminate the parameter to find a Cartesian equation of the curve. Also, sketch the curve and indicate with an arrow the direction in which the curve is traced as the parameter increases $x = e^{t} - 1$, $y = e^{2t}$

Q.2 Find the tangent to the cycloid $x = r(\theta - \sin\theta)$, $y = r(1 - \cos\theta)$ at the point where $\theta = \frac{\pi}{3}$. Also, at what points is the tangent horizontal? When is it vertical?